

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptron's, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

Assessment Tool Weightage: 10%

QUESTIONS:15

Total Marks:100

Annexure A

DEBUGGING & OPTIMIZATION TASK QUESTIONS

Code of Conduct:

I D.RITHVIK (192312119)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of
the student:
Total Marks
Awarded**

Annexure A

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	30%	20%	30%	10%	10%	
Q1						
Q2						
Q3						

Short Answer Test

1. A perceptron model is trained on a dataset that is known to be linearly separable, yet the algorithm fails to converge even after many iterations. Analyze the possible causes of this issue, including improper learning rate selection, incorrect weight initialization, bias handling errors, or flawed implementation of the update rule. Propose debugging strategies to identify the root cause and suggest optimization techniques to ensure faster and stable convergence. (5 Marks)

2. While training a multilayer perceptron using the backpropagation algorithm, the network exhibits slow convergence and gets stuck in local minima. Examine the potential reasons such as vanishing gradients, inappropriate activation functions, poor weight initialization, or suboptimal learning rates. Describe step-by-step

debugging approaches to trace the issue and recommend optimization techniques like adaptive learning rates, normalization, or advanced optimizers to improve performance. (5 Marks)

3. A genetic algorithm designed for hypothesis space search is producing inconsistent and suboptimal solutions across multiple runs. Investigate possible issues such as improper encoding of chromosomes, incorrect fitness function design, premature convergence, or lack of diversity in the population. Explain how to debug these problems systematically and suggest optimization strategies like mutation tuning, selection pressure adjustment, and elitism to enhance solution quality and convergence speed. (5 Marks)

4. A deep neural network combined with genetic programming is used for solving a complex prediction problem, but it suffers from overfitting and high computational cost. Identify debugging steps to analyze issues related to model complexity, over-parameterization, and insufficient training data. Propose optimization techniques such as regularization, dropout, pruning, and efficient evaluation models to balance accuracy and computational efficiency while improving generalization capability. (5 Marks)

5. A decision tree model trained on a large dataset shows high accuracy on training data but poor performance on unseen data. Analyze possible causes such as overfitting, deep tree structure, or noisy data. Propose debugging steps and optimization methods like pruning, cross-validation, and feature selection to improve generalization. (5 Marks)

6. A convolutional neural network (CNN) used for image classification produces unstable results across training epochs. Examine potential issues such as improper data augmentation, vanishing gradients, or batch normalization errors. Describe debugging strategies and suggest optimization techniques to stabilize training and improve accuracy. (5 Marks)

7. A reinforcement learning agent fails to learn an optimal policy in a simulated environment. Investigate causes such as poor reward design, insufficient exploration, or incorrect state representation. Explain debugging methods and recommend improvements like reward shaping, exploration strategies, and tuning hyperparameters. (5 Marks)

8. A clustering algorithm like K-means produces inconsistent cluster assignments for similar datasets. Analyze possible reasons such as poor initialization, incorrect number of clusters, or sensitivity to outliers. Propose debugging approaches and optimization strategies like K-means++, normalization, and silhouette analysis. (5 Marks)

9. A support vector machine (SVM) model underperforms on a nonlinear dataset. Identify issues such as incorrect kernel choice, improper parameter tuning, or scaling problems. Suggest debugging techniques and optimization methods including kernel selection, hyperparameter tuning, and feature scaling. (5 Marks)

10. A natural language processing (NLP) model using recurrent neural networks (RNNs) suffers from poor long-term dependency learning. Examine causes such as vanishing gradients and limited memory capacity. Describe debugging steps and propose solutions like LSTM/GRU units, attention mechanisms, and improved training strategies. (5 Marks)

11. A support vector machine (SVM) model trained for classification is producing very low accuracy on both training and test datasets. Analyze possible causes such as improper kernel selection, incorrect hyperparameter tuning (C and gamma), or poor feature scaling. Describe systematic debugging steps to identify the issue and suggest optimization techniques like kernel tuning, normalization, and grid search to improve model performance. (5 Marks)

12. A K-Means clustering algorithm is applied to a dataset, but the clusters formed are highly imbalanced and do not reflect the actual data distribution. Investigate possible issues such as incorrect choice of K, poor initialization of centroids, or presence of outliers. Explain debugging approaches and suggest improvements like the Elbow method, K-Means++, and outlier removal techniques. (5 Marks)

13. A recurrent neural network (RNN) designed for time-series prediction shows inconsistent outputs and fails to capture long-term dependencies. Analyze possible causes such as vanishing gradients, insufficient sequence length, or improper architecture design. Describe debugging steps and propose optimization techniques such as using LSTM/GRU units, gradient clipping, and sequence normalization. (5 Marks)

14. A Naïve Bayes classifier used for text classification performs poorly when applied to real-world data containing noisy and irrelevant features. Examine possible issues such as violation of feature independence assumption, improper preprocessing, or imbalanced dataset. Explain debugging strategies and suggest improvements like feature selection, smoothing techniques, and data balancing methods. (5 Marks)

15. A reinforcement learning agent trained to solve a navigation problem fails to learn an optimal policy and shows erratic behavior. Analyze possible causes such as sparse rewards, improper exploration-exploitation balance, or incorrect reward function design. Describe debugging approaches and recommend optimization strategies like reward shaping, epsilon decay tuning, and policy optimization methods to improve learning performance. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	30	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	10	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Perceptron fails to converge although data is linearly separable

Possible Causes:

- Learning rate is too high or too low.
- Incorrect weight initialization.
- Bias term is missing or updated incorrectly.
- Wrong implementation of the perceptron update rule.

Debugging Steps:

- Verify the update rule: $W = W + \eta(y - \hat{y})x$
- Check whether bias is updated correctly.
- Monitor weight values after each iteration.
- Test the algorithm on a small linearly separable dataset.

Optimization Techniques:

- Choose an appropriate learning rate.
- Normalize input features.
- Initialize weights with small random values.
- Shuffle training samples before each epoch.

2. MLP with Backpropagation shows slow convergence

Possible Causes:

- Vanishing gradients.
- Poor weight initialization.
- Inappropriate activation function (e.g., sigmoid).
- Incorrect learning rate.

Debugging Steps:

- Monitor training loss and gradients.
- Check weight updates after every epoch.
- Compare different activation functions.
- Verify backpropagation calculations.

Optimization Techniques:

- Use ReLU activation.
- Apply Xavier/He initialization.
- Use Adam optimizer.
- Normalize input data.

3. Genetic Algorithm produces inconsistent solutions

Possible Causes:

- Poor chromosome encoding.
- Incorrect fitness function.
- Premature convergence.
- Low population diversity.

Debugging Steps:

- Validate chromosome representation.
- Test the fitness function separately.
- Monitor crossover and mutation operations.
- Track population diversity.

Optimization Techniques:

- Increase mutation probability.
- Adjust selection pressure.
- Apply elitism.
- Increase population size.

4. Deep Neural Network with Genetic Programming suffers from overfitting

Possible Causes:

- Model is too complex.
- Too many parameters.
- Small training dataset.
- High computational cost.

Debugging Steps:

- Compare training and testing accuracy.
- Analyze model complexity.
- Monitor validation loss.

Optimization Techniques:

- Apply Dropout.
- Use L1/L2 Regularization.
- Perform model pruning.
- Use Early Stopping.

5. Decision Tree performs poorly on unseen data

Possible Causes:

- Overfitting.
- Very deep tree.
- Noisy or irrelevant features.

Debugging Steps:

- Compare training and testing accuracy.
- Inspect tree depth.
- Check feature importance.

Optimization Techniques:

- Apply pruning.
- Use cross-validation.
- Select important features.
- Limit tree depth.

6. CNN produces unstable results during training

Possible Causes:

- Improper data augmentation.
- Vanishing gradients.
- Batch normalization errors.
- High learning rate.

Debugging Steps:

- Monitor loss and accuracy.
- Verify augmented images.
- Check gradient values.
- Validate Batch Normalization layers.

Optimization Techniques:

- Use proper augmentation.
- Apply Batch Normalization.
- Use learning-rate scheduling.
- Use Adam optimizer.
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7. Reinforcement Learning agent fails to learn optimal policy

Possible Causes:

- Poor reward function.
- Insufficient exploration.
- Incorrect state representation.

Debugging Steps:

- Monitor rewards.
- Observe agent actions.
- Verify state transitions.

Optimization Techniques:

- Apply reward shaping.
- Use ϵ -greedy exploration.
- Tune learning rate and discount factor.
- Improve state representation.

8. K-Means produces inconsistent clusters

Possible Causes:

- Random centroid initialization.
- Incorrect number of clusters.
- Presence of outliers.

Debugging Steps:

- Visualize clusters.
- Check centroid positions.
- Calculate inertia.

Optimization Techniques:

- Use K-Means++.
- Normalize features.
- Remove outliers.
- Use Silhouette Score to choose K.

9. SVM underperforms on nonlinear data

Possible Causes:

- Incorrect kernel.
- Improper C and Gamma values.
- Unscaled features.

Debugging Steps:

- Compare different kernels.
- Check feature scaling.
- Evaluate validation accuracy.

Optimization Techniques:

- Use RBF kernel.

- Perform Grid Search.
- Normalize features.
- Tune C and Gamma.

10. RNN fails to learn long-term dependencies

Possible Causes:

- Vanishing gradients.
- Limited memory.
- Long sequences.

Debugging Steps:

- Monitor gradients.
- Check sequence outputs.
- Evaluate training loss.

Optimization Techniques:

- Replace RNN with LSTM/GRU.
- Use Attention Mechanism.
- Apply Gradient Clipping.
- Normalize input sequences.

11. SVM has very low training and testing accuracy

Possible Causes:

- Wrong kernel selection.
- Poor C and Gamma values.
- Unscaled features.

Debugging Steps:

- Check confusion matrix.
- Evaluate feature scaling.
- Compare different kernels.

Optimization Techniques:

- Perform Grid Search.
- Normalize features.
- Select suitable kernel.
- Use cross-validation.

12. K-Means forms imbalanced clusters

Possible Causes:

- Incorrect value of K.
- Poor centroid initialization.
- Presence of outliers.

Debugging Steps:

- Inspect cluster sizes.
- Analyze centroid locations.
- Visualize data.

Optimization Techniques:

- Apply Elbow Method.
- Use K-Means++.
- Remove outliers.
- Normalize data.

13. RNN for time-series prediction gives inconsistent outputs

Possible Causes:

- Vanishing gradients.
- Short sequence length.
- Poor architecture.

Debugging Steps:

- Monitor gradient values.
- Check prediction errors.
- Validate sequence inputs.

Optimization Techniques:

- Use LSTM/GRU.
- Apply Gradient Clipping.
- Normalize sequences.
- Tune learning rate.

14. Naïve Bayes performs poorly on noisy text data

Possible Causes:

- Violation of feature independence.
- Irrelevant features.
- Imbalanced dataset.

Debugging Steps:

- Inspect preprocessing.
- Analyze feature importance.
- Check class distribution.

Optimization Techniques:

- Perform feature selection.
- Apply Laplace smoothing.
- Balance the dataset.
- Remove noisy features.

15. Reinforcement Learning navigation agent behaves erratically

Possible Causes:

- Sparse rewards.
- Poor exploration strategy.
- Incorrect reward function.

Debugging Steps:

- Monitor rewards and actions.
- Observe policy updates.
- Verify environment transitions.

Optimization Techniques:

- Apply reward shaping.
- Tune epsilon decay.
- Use policy optimization methods (PPO/DQN).
- Adjust learning rate and discount factor.